4. DUALITY LINEAR PROGRAMMING

MBS

NUMERICAL QUESTIONS

1. 2070 Q.No. 11

Write the dual of the following problem and obtain the optimal solution of the dual problem by using simplex method.

Minimize $C = Rs.300X_1 + 200X_2 + 400X_3$

Subject to the constraint

$$3X_1 + 2X_2 + X_3 \geq 2$$
$$4X_1 + X_2 + 3X_3 \geq 4$$
$$2X_1 + 2X_2 + 2X_3 \geq 3$$

Where $X_1, X_2, X_3 \geq 0$ non negative condition. [20]

Ans: Primal Solution: Min $Z = \frac{1,150}{3}$; $X_1 = \frac{5}{6}$, $X_2 \frac{2}{3}$, $X_3 = 0$; Dual Solution: Max $Z = \frac{1,150}{3}$; $Y_1 = 0$, $Y_2 = \frac{100}{3}$ $Y_3 = \frac{250}{3}$

2. 2069 Q. No. 10

Write the dual of the problem and convert inequality into equality by adding slack of surplus variables. [10]

Min. $Z = 2x_1 + 3x_2 + x_3$

S.t. $4x_1 + 3x_2 + x_3 = 6$

$x_1 + 2x_2 + 5x_3 = 4$

$x_1, x_2, x_3 \geq 0.$

Ans: Max $Z = -6y_1 + 6y_2 - 4y_3 + 4y_4$; Subject to: $-4y_1 + 4y_2 - y_3 + y_4 + S_1 = 2$; $-3y_1 + 3y_2 - 2y_3 + 2y_4 + S_2 = 3$; $-y_1 + y_2 - 5y_3 + 5y_4 + S_3 = 1$; $y_1, y_2, y_3, y_4, S_1, S_2, S_3 \geq 0$

3. 2067 Q.No. 11

Write the dual of the following problem and then solve the dual problem by using simplex method. Finally interpret the primal and dual values from the dual solution. [20]

Maximize profit $Z =$ Rs. $4,000\ P_1 +$ Rs. $3,200\ P_2$

Subject to the constraints

Raw material (kg) $20P_1 + 10P_2 \leq 300$

Labour in assembly (man – hours) $2P_1 + 5P_2 \leq 50$

Labour in finishing (man – hours) $2P_1 + 3P_2 \leq 38$

Where P_1 and P_2 are two different products and $P_1 P_2 \geq 0$.

Ans: Dual solution: Min Z = 64,800; $Y_1 = 140$; $Y_2 = 0$; $Y_3 = 600$; Primal solution: Max Z = 64,800; $P_1 = 13$; $P_2 = 4$

4. **2062 Q.No. 7**

Find the optimum solution for the following problem by using simplex.

Minimize the cost = Rs. 300A + Rs. 800B

Subject to:

$A + B = 200$ $\qquad A \leq 40$ $\qquad B \geq 30$ $\qquad$ where $A, B \geq 0$

Interpret the value of dual from the final table of the primal problem. [20]

Ans: A = 40; B = 160, Minimum cost = Rs. 1,40,000 dual variable Y_1 = Rs. 800; Y_2 = Rs. 500, and Y_3 = Rs. 0, maximum profit = Rs. 1,40,000

5. **2061 Q.No. 7**

Write the dual of the following problem and solve them dual problem by using simplex method and interpret the result.

Maximize the profit Rs. $16X_1 + 10X_2$

Subject to:

$4X_1 + 2X_2 \leq 24$ $\qquad 3X_1 + 3X_2 \leq 21$ $\qquad 2X_1 + 5X_2 \leq 30$ $\qquad$ where, $X_1 \geq 0$ and $X_2 \geq 0$ [20]

Ans: Min (c) = Rs. 100, $W_1 = 3$, $W_2 = 4/3$

6. **2058 Q.No. 7**

Obtain the dual of the following problem and interpret the result by finding the solution of the dual problem by using simplex method.

Maximize the profit: Rs. $400X_1 + 320X_2$

Subject to: Assembly constraint in hour $40X_1 + 20X_2 \leq 600$

Product cost constraint in rupees $40X_1 + 100X_2 \leq 1000$

Finishing constraint in hour $2X_1 + 3X_2 \leq 38$ $\qquad$ where, $X_1 \geq 0$ and $X_2 \geq 0$ [20]

Ans: Min (c) = Rs. 6480, $W_1 = 7$, $W_2 = 0$, $W_3 = 60$

M-B.A

7. **2053 Q.No. 6**

Write the dual of the given LPP and obtain its solution by using simplex method. From the dual solution also read the solution of primal LP?

Minimize C = Rs. $(60X_1 + 40X_2 + 80X_3)$

Subject to

$3X_1 + 2X_2 + X_3 \geq 2$ $\qquad 4X_1 + X_2 + 3X_3 \geq 4$

$2X_1 + 2X_2 + 2X_3 \geq 3$ $\qquad$ and $X_1, X_2, X_3 \geq 0$

Ans: $X_1 = 5/6$; $X_2 = 2/3$; $X_3 = 0$, Cost (c) = Rs. 230/3